

Tide retrospective: the quadratic bridge and Gaussian absorption the domain layer of the numerator stage

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Abstract

The forward-expansion domain ties the matrix H to the loss only through the quadratic Peano condition, and its exponent split speaks of diagonal Taylor terms, so the identification $T_2 = \frac{1}{2}$ qform H has to be *proven*, not assumed. The proof is a pleasing self-consumption: compose both Peano statements along the ray $t \mapsto t \cdot z$, view each as an order-2 asymptotic polynomial for $t \mapsto L(tz)$, and apply the programme’s own stage-1 coefficient uniqueness. At index one this yields the vanishing of the degree-one diagonal term — the critical-point condition that the exponent split takes as a hypothesis, now discharged for every domain — and at index two the bridge itself. With the bridge in hand, the domain’s coercivity constant gives the named lower bound $\frac{\lambda}{2}\|z\|^2 \leq T_2(z)$, and the mesoscopic scale arithmetic ($q^s\|z\|^{s+2} = \|z\|^2(q\|z\|)^s \leq \|z\|^2\sqrt{q}$) bounds every correction by $\varepsilon\|z\|^2$ on the window, absorbing the total absolute correction into a Gaussian at a quarter of the rate: on the window, eventually,

$$e^{-T_2(z)} e^{\sum_s q^s |V_s(z)| + |q^N \rho_q(z)|} \leq e^{-(\lambda/4)\|z\|^2}.$$

1 Setting

Stage 5c-pre of the forward-expansion programme: items 1–3 of the archived architecture consult’s implementation order, the layer where domain-specific uniformity concentrates before the single q -uniform majorant. The consult specifically recommended the sum-of-absolute-values form for the correction bound (the scalar estimates use $e^{|A|+|\delta|}$, so bounding the sum of absolute values avoids cancellation questions), and bridging T_2 to the quadratic form *first* so the integration layer never rewrites between the two languages.

2 The thing formalised

The ray expansions `rayExpansion_taylor` and `rayExpansion_quad`: the loss along a ray is an order-2 asymptotic polynomial with coefficients the diagonal Taylor terms (peel degrees ≥ 3 as $o(t^2)$, the Peano remainder transported along the ray) or the quadratic-form coefficients (the domain’s quadratic Peano along the same ray). Uniqueness (`isAsymptoticExpansionTo_coeff_eq`, stage A1) gives `taylorHomogeneousTerm_one_eq_zero` and `taylorHomogeneousTerm_two_eq_qform`;

`t2_lower` then carries the domain's λ . Window control: `eventually_abs_scaledRem_le` extracts the little- o epsilon-delta from `taylorPeano` and routes window points into the δ -ball via the stage-2 lemma; `eventually_abs_scaledPerturbation_le_quadratic` converts order $N+2$ to quadratic through $(q\|z\|)^N \leq 1$; `eventually_exponentCorrection_le_quadratic` runs the $\sqrt{q}$ scale arithmetic against the operator-norm bounds; and `gaussian_absorb` combines them at $\varepsilon = \lambda/8$ each. The reusable helper `integrable_one_add_norm_pow_mul_gaussian` closes the file (binomial expansion against the seabed's polynomial-Gaussian integrability).

3 Where progress stalled

Barely. `Real.sqrt_le_one` is an iff, not an implication; λ cannot appear inside an identifier (`hλ` fails to parse — it is the anonymous-function token); and an unused domain binder tripped the zero-warnings gate, fixed by the underscore convention with dot-notation callers unaffected.

4 Mathlib lookup versus new mathematics

Lookup: `Metric.eventually_nhds_iff`, `Real.mul_self_sqrt`, `Real.sqrt_le_one`, `pow_le_one0`, `pow_le_pow_left0`, `add_pow`, `integrable_finset_sum`, `Asymptotics.IsLittleO.sum`. New: no new mathematics; the structural point is the self-consumption — the programme's stage-1 uniqueness closing the programme's own central identification — and the observation that the critical-point condition is derivable rather than hypothesised.

5 What the next tide inherits

Stage 5c proper: define the rescaled numerator, prove the normalized q -uniform Gaussian majorant on the window (this tide's absorption $\times$ 5b's three scalar bounds $\times$ 5a's coefficient growth), apply the filter-indexed dominated convergence theorem with the window indicator folded into the integrand, remove the two outer tails with the stage-2 calculus, and package `numerator_hasExpansion`. The exponent split's hypothesis should now be discharged with `taylorHomogeneousTerm_one_eq_zero` at every call site.